

Nicholas Krause

3922 W 126th Ave – Broomfield, CO

☎ (720) 957 6837 • 🌐 github.com/nicholas-krause

✉ nicholask.12210@gmail.com

Summary

Software engineer with more than five years of experience in feature development and a strong passion for researching and leveraging emerging technologies.

Experience

Embedded Linux Firmware Engineer – *Trimble Inc., Westminster, CO* *Jul 2023 – Present*

- Developed and maintained embedded Linux firmware using Yocto Project, BitBake, and Python in Ubuntu-based environments.
- Designed and implemented an OTA firmware update solution, integrating AsyncAPI and RESTful APIs for seamless update delivery.
- Played a key role in transitioning the CI/CD pipeline from Jenkins to Bamboo, enabling more efficient automated build and test workflows, leveraging the OTA firmware system for validation.
- Containerized the embedded OS using Yocto meta-virtualization layer and created OCI-compliant images with Podman for on-device application deployment.

Rotational Development Program – *Trimble Inc., Christchurch, NZ* *Jul 2021 – Jul 2023*

Completed four 6-month rotations of a software engineering development track, gaining hands-on experience across embedded systems, web development, and RD initiatives:

- Embedded Android Development (Earthworks) – Developed new features in Java; contributed to central unit testing using Go and TypeScript.
- Front-End Web Development (WorksOS) – Delivered production-ready features using Angular; created Azure pipelines; wrote unit tests in TypeScript.
- Embedded C++ Development (Autonomous Solutions) – Developed ROS nodes and unit tests; contributed to early-stage hardware/software integration; utilized Docker and Linux/Ubuntu environments.
- RD for Joint Venture – Conducted feasibility research and prototyping in C++ for emerging technologies; assessed risks and integration potential.

Software Engineer – *Trimble Inc., Christchurch, NZ* *Dec 2020 – Jul 2021*

- Contributed to the development of a cross-platform mobile application for Windows and Android using C++ and Qt libraries

Graduate Controls Engineer – *Scott Technology, Christchurch, NZ* *Dec 2019 – Jun 2020*

- Developed foundational controls engineering skills through hands-on work and mentorship from senior engineers. Designed electrical schematics using EPLAN and assisted in wiring electrical cabinets for automation systems.

Skills

Languages: Python, C/C++, Java, Typescript

Technologies: Yocto project, CI/CD pipelines(Jenkins/Bamboo), Automated testing(Slash/Robot framework), ROS framework, AndroidOS, Git, Bash, podman/Docker, Angular

Education

Bachelors of Mechatronics Engineering(with Honors)

University of Canterbury

First Class Honours

Otago Boy's High School

Christchurch, NZ

2016–2020

Dunedin, NZ

2011–2015